SYSTEM ARCHITECTURE
& WORKFLOW DOCUMENTATION

Multi-Role Super Platform

Platform	Super App + Fintech + Marketplace
Modules	11 Core Modules
User Roles	8 Role Types (RBAC)
Build Phases	7 Phases — Developer Ready
Architecture	Modular • Config-Driven • Multi-Country

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1. System Overview

This platform is a comprehensive Multi-Role Super App combining services, digital wallet, job marketplace, agent network, and multi-country administrative control into a single unified system.

1.1 Platform Identity

Attribute	Description
Type	Super App + Fintech + Marketplace Hybrid
Architecture	Modular Microservices — Independently deployable modules
Design Principle	Config-driven — No hardcoded business logic
Scale	Multi-country, Multi-currency, Multi-language ready
Access Control	Role-Based Access Control (RBAC) — 8 distinct roles
Financial Safety	Transaction-safe — Atomic operations, Idempotency keys

1.2 Key Design Principles

- Modular system — each module independently deployable and scalable
- Config-driven — all business rules stored in DB, no hardcoding
- Multi-country ready — services toggled per country/region
- Role-based architecture — 8 roles with granular permission control
- Transaction-safe — atomic wallet operations with rollback support
- Event-driven commission — async processing via message queues

2. User Roles & Access Control

The platform supports 8 distinct user roles organized in a hierarchical RBAC system. Each role has a clearly defined scope of access and capabilities.

Role	Category	Key Capability	Access Rights
User	Base	Service access only	Browse & use services (recharge, bills). No wallet.
Member	Financial	Wallet + All services	Full services + Wallet after KYC verification.
Saathi	Agent	Tasks + Commission	Accept tasks from Users, complete them, earn commission.
Business	Enterprise	Jobs + Ads	Post jobs, run ads, manage business profile and hiring.
District Partner	Hierarchy	District monitoring	Monitor and control activity within assigned district.
State Partner	Hierarchy	State-level control	Oversee multiple districts, state-level reporting.
Country Head	Hierarchy	Country service control	Enable/disable services for the entire country.
Admin / Sub-Admin	Super	Full system control	All modules: users, finance, analytics, configuration, partners.

2.1 Role Upgrade Path

Start		Action Required		Result
Register (phone/email)	→	Verify OTP / complete profile	→	**User role activated**
User with basic account	→	Submit KYC documents	→	**Member — Wallet created**
Member in good standing	→	Apply for Saathi role	→	**Saathi — Task + Commission access**

3. Core Workflows

This section describes the primary workflows for each user type. Developers should implement these flows in the order shown to ensure correct dependencies.

3.1 User Registration & Onboarding Flow

Step	Actor	Action	System Response
1	User	Opens app, enters phone/email	Send OTP to phone/email
2	User	Enters OTP code	Verify OTP, create user record with role=USER
3	User	Completes profile (name, DOB, address)	Profile saved, JWT token issued
4	User	Requests Member upgrade	Prompt KYC document upload
5	User	Uploads KYC documents (Aadhaar/PAN)	Documents stored, status=PENDING
6	Admin/System	Reviews KYC documents	Approve or reject KYC
7	System	KYC approved	role upgraded to MEMBER, Wallet created automatically
8	Member	Views wallet	Wallet shows INR 0.00 balance, ready for use

3.2 Wallet & Transaction Flow

Step	Actor	Action	System Response
1	Member	Initiates Add Money (amount + payment method)	Redirect to payment gateway
2	Payment Gateway	Processes payment	Sends success/failure callback
3	System	Receives success callback	Generate idempotency key, lock wallet row
4	System	Create transaction record	Insert txn (status=PROCESSING)
5	System	Update wallet balance	Atomic UPDATE balance WHERE id=? AND version=?
6	System	Mark transaction complete	txn status=COMPLETED, release lock
7	Member	Views updated balance	Real-time balance reflects new amount
8	Member	Initiates Send Money	Validate recipient, check balance
9	System	Process transfer	Debit sender, credit receiver atomically

10	System	Distribute commission	Async queue: calculate & pay Saathi/Partner shares

3.3 Saathi Task Flow

Step	Actor	Action	System Response
1	User	Creates task (type, location, details, budget)	Task saved, status=PENDING
2	System	Finds nearby Saathi agents	Notify Saathi via push/SMS (2km radius first)
3	Saathi	Views task notification	Task details shown with Accept/Decline options
4	Saathi	Accepts the task	Task status=ACCEPTED, User notified
5	Saathi	Travels to location, starts task	Status=IN_PROGRESS
6	Saathi	Marks task complete with proof (photo)	Status=COMPLETED, User notified
7	User	Confirms task completion	Status=CONFIRMED
8	System	Releases payment from escrow	Saathi wallet credited
9	System	Distributes commission	District→State→Country→Admin shares paid async
10	User	Rates Saathi (1-5 stars)	Rating recorded, affects Saathi ranking
11	System	If no Saathi accepts in 15 min	Auto-reassign: expand radius to 5km, re-notify

3.4 Business User Flow (Jobs & Ads)

Step	Actor	Action	System Response
1	Business	Registers business account	Business profile created, status=PENDING_VERIFICATION
2	Admin	Verifies business documents	Business status=VERIFIED, full access granted
3	Business	Posts a Job (title, desc, location, salary)	Job saved, status=PENDING_APPROVAL
4	Admin	Reviews job post	Approves/rejects — if approved, Job goes LIVE
5	User/Saathi	Searches and applies for job	Application record created
6	Business	Reviews applications	Shortlists, schedules interviews
7	Business	Creates an Ad campaign	Set budget, target region/role, schedule
8	System	Validates ad budget	Checks business wallet balance
9	System	Runs ad as scheduled	Deducts cost per impression/click from wallet
10	Business	Views ad analytics	Impressions, clicks, conversions dashboard

3.5 Partner Hierarchy & Country Control Flow

Step	Actor	Action	System Response
1	Admin	Creates Country in system	Country record created with config_json
2	Admin	Assigns Country Head user	User role upgraded to COUNTRY_HEAD
3	Country Head	Enables specific services for country	service_config table updated for that country
4	Country Head	Assigns State Partners	State Partner records created under country
5	State Partner	Assigns District Partners	District Partner records under state
6	User	Opens app in that country	Only enabled services are visible
7	System	Any transaction in country	Commission distributed: Saathi→District→State→Country Head→Admin
8	Country Head	Views country analytics	Revenue, user count, top services, partner earnings

4. Core Modules — Developer Reference

Each module is an independent microservice. Modules communicate via REST APIs internally and event queues for async operations.

Module 1: Auth Service

Central authentication and authorization gateway for all services.

API Endpoints	POST /auth/login | POST /auth/register | POST /auth/otp/verify | POST /auth/refresh | POST /auth/logout
Dev Notes	All other services validate JWT from this service. Role permissions cached in Redis. 2FA/OTP via SMS gateway.

Module 2: User & Member Service

Manages user profiles, membership upgrade, and KYC processing.

API Endpoints	GET /users/:id | PUT /users/:id | POST /users/kyc/submit | GET /users/kyc/status | POST /users/upgrade-member
Dev Notes	KYC document stored in object storage (S3/GCS). On KYC approval, emits MEMBER_APPROVED event consumed by Wallet Service.

Module 3: Wallet Service

Handles digital wallet creation, balance management, and fund transfers.

API Endpoints	POST /wallet/add-money | POST /wallet/transfer | POST /wallet/withdraw | GET /wallet/balance | GET /wallet/history
Dev Notes	Wallet only created for MEMBER+ roles. Uses optimistic locking (version column) on balance updates. All transfers are atomic.

Module 4: Service Engine

Config-driven utility services: mobile recharge, DTH, bill payments.

API Endpoints	GET /services | POST /services/recharge | POST /services/bill-pay | GET /services/history | GET /services/providers
Dev Notes	Services list fetched from service_config JOIN country_config. Integrates with BBPS and operator APIs. Commission triggered on each successful service transaction.

Module 5: Task Service

Core task booking and Saathi assignment engine.

API Endpoints	POST /tasks | GET /tasks/:id | PATCH /tasks/:id/status | GET /tasks/nearby | POST /tasks/:id/rate
Dev Notes	Assignment uses geo-query (PostGIS). Auto-escalation via cron job after 15 min timeout. Task amount held in escrow until completion.

Module 6: Commission Service

Multi-level commission calculation and distribution engine.

API Endpoints	POST /commission/calculate | GET /commission/ledger | GET /commission/earnings/:userId
Dev Notes	Event-driven via message queue (RabbitMQ/Kafka). commission_rules table controls % per level per country. Append-only ledger — no updates, only inserts.

Module 7: Jobs & Ads Service

Business marketplace for job postings and advertisement campaigns.

API Endpoints	POST /jobs | GET /jobs | POST /jobs/:id/apply | POST /ads/campaign | GET /ads/analytics
Dev Notes	All posts require Admin approval before going live. Ad billing deducted from Business wallet in real-time. Targeting by country, state, district, and role type.

Module 8: Partner Service

Multi-level partner hierarchy management and monitoring.

API Endpoints	POST /partners/assign | GET /partners/hierarchy | GET /partners/analytics | GET /partners/earnings
Dev Notes	Hierarchy: Admin → Country Head → State Partner → District Partner. Each level only sees downward data. Commission aggregated up the hierarchy.

Module 9: Country Config Service

Country-specific service visibility and configuration management.

API Endpoints	POST /country | PUT /country/:id/services | GET /country/:id/config | GET /services/available
Dev Notes	All service visibility queries JOIN this table. Stores currency, KYC rules, enabled services as JSON config per country.

Module 10: Admin Service

Full system control, analytics, and administrative operations.

API Endpoints	GET /admin/dashboard | GET /admin/users | PUT /admin/users/:id/status | GET /admin/reports | POST /admin/config
Dev Notes	Sub-Admin roles have module-level RBAC. Real-time stats via WebSocket. Audit log records all admin actions with actor, timestamp, and IP.

5. Database Schema Reference

Core database entities. All tables include created_at, updated_at timestamps. Use UUID for primary keys for distributed system compatibility.

users

Column	Type	Constraint	Notes
id	UUID	PK	Unique user identifier
phone	VARCHAR(15)	UNIQUE NOT NULL	Primary login identifier
email	VARCHAR(255)	UNIQUE	Optional secondary identifier
role	ENUM	NOT NULL	USER, MEMBER, SAATHI, BUSINESS, DISTRICT, STATE, COUNTRY, ADMIN
status	ENUM	NOT NULL	ACTIVE, SUSPENDED, PENDING
country_id	UUID	FK → countries	User's registered country
full_name	VARCHAR(255)	NOT NULL	Display name
created_at	TIMESTAMP	NOT NULL	Auto-set on insert

kyc

Column	Type	Constraint	Notes
id	UUID	PK	
user_id	UUID	FK → users	One KYC per user
doc_type	ENUM	NOT NULL	AADHAAR, PAN, PASSPORT, NATIONAL_ID
doc_number	VARCHAR(50)	NOT NULL	Encrypted at rest
doc_url	TEXT	NOT NULL	Object storage URL
status	ENUM	NOT NULL	PENDING, APPROVED, REJECTED
verified_by	UUID	FK → users (admin)	Admin who verified
verified_at	TIMESTAMP		Null until verified

wallets

Column	Type	Constraint	Notes
id	UUID	PK	
user_id	UUID	FK → users UNIQUE	One wallet per Member

balance	DECIMAL(15,2)	NOT NULL DEFAULT 0	Current balance
currency	VARCHAR(3)	NOT NULL	INR, USD, etc.
status	ENUM	NOT NULL	ACTIVE, FROZEN, CLOSED
version	INTEGER	NOT NULL DEFAULT 0	Optimistic lock counter

transactions

Column	Type	Constraint	Notes
id	UUID	PK	
wallet_id	UUID	FK → wallets	Owner wallet
type	ENUM	NOT NULL	CREDIT, DEBIT, TRANSFER, REFUND
amount	DECIMAL(15,2)	NOT NULL	Always positive
ref_id	VARCHAR(100)	UNIQUE	Idempotency key from payment gateway
status	ENUM	NOT NULL	PROCESSING, COMPLETED, FAILED, REVERSED
meta_json	JSONB		Provider response, task_id, etc.
created_at	TIMESTAMP	NOT NULL INDEXED	Partition key for monthly sharding

tasks

Column	Type	Constraint	Notes
id	UUID	PK	
user_id	UUID	FK → users	Task creator
saathi_id	UUID	FK → users NULLABLE	Null until accepted
type	VARCHAR(50)	NOT NULL	Task category code
status	ENUM	NOT NULL	PENDING, ACCEPTED, IN_PROGRESS, COMPLETED, CONFIRMED, CANCELLED
amount	DECIMAL(10,2)	NOT NULL	Task payment amount
location_lat	DECIMAL(10,8)		PostGIS indexed
location_lng	DECIMAL(11,8)		PostGIS indexed
escrow_txn_id	UUID	FK → transactions	Held until task confirmed

commission_rules

Column	Type	Constraint	Notes
id	UUID	PK	
level	ENUM	NOT NULL	SAATHI, DISTRICT, STATE, COUNTRY, ADMIN
trigger_type	ENUM	NOT NULL	TASK, SERVICE, TRANSFER
percent	DECIMAL(5,2)	NOT NULL	Commission percentage 0-100
country_id	UUID	FK → countries	Country-specific rules
active	BOOLEAN	NOT NULL DEFAULT TRUE	Soft enable/disable

commission_ledger

Column	Type	Constraint	Notes
id	UUID	PK	
trigger_type	ENUM	NOT NULL	TASK, SERVICE, TRANSFER
trigger_id	UUID	NOT NULL	FK to task/transaction
user_id	UUID	FK → users	Who earns this commission
amount	DECIMAL(10,2)	NOT NULL	Commission amount paid
level	ENUM	NOT NULL	Which hierarchy level earned
created_at	TIMESTAMP	NOT NULL	Append-only — never UPDATE this table

countries + service_config

Column	Type	Constraint	Notes
countries.id	UUID	PK	
countries.name	VARCHAR(100)	NOT NULL	Country display name
countries.code	CHAR(2)	UNIQUE NOT NULL	ISO alpha-2 code
countries.currency	CHAR(3)	NOT NULL	ISO currency code
service_config.country_id	UUID	FK → countries	Which country
service_config.service_id	VARCHAR(50)	NOT NULL	Service code e.g. MOBILE_RECHARGE
service_config.enabled	BOOLEAN	NOT NULL DEFAULT FALSE	Toggled by Country Head
service_config.config_json	JSONB		Service-specific overrides

6. Build Phases — Developer Roadmap

Phases are ordered by dependency. Each phase produces a working deliverable. Developers can be assigned across phases — Phase 1 must complete before Phase 2 begins.

Phase 1: Auth + User + Member

Timeline	2-3 weeks | 2 developers
Deliverable	Users can register, verify, complete KYC, and upgrade to Member.

- Set up project structure, CI/CD pipeline, Docker environments
- Auth Service: JWT, OTP, refresh tokens, RBAC middleware
- User Service: registration, profile management, role management
- KYC Service: document upload, admin review workflow, status tracking
- Member upgrade: KYC approval triggers MEMBER_APPROVED event
- Unit tests for auth flows, integration tests for registration

Phase 2: Wallet + Transaction System

Timeline	2-3 weeks | 2 developers
Deliverable	Members can add money, transfer funds, and view transaction history.

- Wallet Service: creation on MEMBER_APPROVED event consumption
- Add money flow: payment gateway integration (Razorpay/Stripe)
- Transaction engine: idempotency keys, optimistic locking, atomic updates
- Transfer flow: sender debit + receiver credit in single DB transaction
- Transaction history API with pagination and filtering
- Wallet freeze/unfreeze admin operations

Phase 3: Service Engine (Recharge / Bills)

Timeline	2 weeks | 2 developers
Deliverable	Members can recharge mobile, pay bills — only services enabled for their country are visible.

- Service config table seeding with initial services per country
- Mobile recharge integration (multiple operators)
- DTH and utility bill payment integration (BBPS)
- Service listing API filtered by country_config
- Service transaction records linked to wallet transactions
- Retry logic for failed service API calls

Phase 4: Saathi + Commission System

Timeline	3-4 weeks | 2-3 developers
Deliverable	Users can book tasks. Saathi agents accept, complete, and earn commission. Multi-level distribution works.

- Saathi profile and onboarding flow
- Task creation, geo-based assignment engine (PostGIS)
- Task status machine: PENDING → ACCEPTED → IN_PROGRESS → COMPLETED → CONFIRMED
- Escrow system: hold task amount until User confirms completion
- Commission Service with event-driven processing via message queue
- Commission rules configuration per country per level
- Auto-reassignment cron for unaccepted tasks (15 min timeout)
- Rating system for Saathi after task completion

Phase 5: Business + Jobs + Ads

Timeline	2-3 weeks | 2 developers
Deliverable	Businesses can post jobs, run ads, manage campaigns. Users/Saathi can apply for jobs.

- Business registration and verification flow
- Job posting CRUD with admin approval workflow
- Job application system with status tracking
- Ad campaign creation with budget, targeting, and scheduling
- Ad billing: real-time deduction from Business wallet
- Basic ad analytics: impressions, clicks, conversions

Phase 6: Partner Hierarchy + Country Control

Timeline	2-3 weeks | 2 developers
Deliverable	Multi-country system live. Partners control regions. Services visible only where enabled.

- Partner role assignment flow (District, State, Country Head)
- Country creation and Country Head assignment
- Service enable/disable toggle per country by Country Head
- KYC rules config per country (required documents)
- Partner commission calculation in commission service
- Partner analytics: earnings, downline activity, region stats

Phase 7: Admin Dashboard + Analytics

Timeline	2-3 weeks | 2 developers
Deliverable	Full system visibility and control. All reports available. Sub-admins can be assigned module access.

- Admin dashboard with real-time system metrics (WebSocket)
- User management: view, filter, suspend, verify users
- KYC review queue and approval interface
- Sub-Admin role creation with module-level RBAC
- Financial reports: revenue, commissions, wallet totals
- Audit log system: all admin actions recorded
- Fraud detection: unusual transaction pattern alerts
- System config management panel

7. Critical Business Logic Rules

These rules MUST be enforced in code. They are non-negotiable business requirements.

7.1 Wallet Access Rules

Rule	Valid	Invalid
Wallet creation	Only after KYC APPROVED	Cannot create for USER role
Wallet access	MEMBER role and above	USER role — 403 Forbidden
Concurrent transactions	Optimistic lock (version column)	Direct UPDATE without version check
Duplicate payment	Idempotency key blocks duplicate	Processing same ref_id twice

7.2 Commission Distribution Rules

- Commission is calculated as a percentage of the transaction/task amount
- Percentages are fetched from commission_rules table — never hardcoded
- Commission is distributed asynchronously via message queue (not in the transaction flow)
- If a hierarchy level has no assigned partner, that commission goes to Admin
- Commission ledger is append-only — entries are NEVER updated or deleted
- Commission is only paid on COMPLETED + CONFIRMED transactions/tasks

7.3 Service Visibility Rules

- Services are only shown to users if service_config.enabled = TRUE for their country
- Service list API MUST JOIN with country_config on every request — no caching of service lists without TTL
- Country Head can enable/disable services at any time — changes take effect immediately
- If a user changes country, their service list is refreshed on next app load

7.4 Task Assignment Rules

- Task amount is held in escrow (separate wallet hold) the moment the task is created
- Escrow is released to Saathi wallet only after User confirms completion (status = CONFIRMED)
- If User does not confirm within 48 hours after COMPLETED, auto-confirm triggers
- If no Saathi accepts within 15 minutes: expand search radius from 2km to 5km, notify again
- If no Saathi accepts within 30 minutes: task moved to OPEN pool, notify all Saathi in district
- Task cancellation by User before acceptance: full refund, no commission
- Task cancellation by User after Saathi acceptance: partial charge (10%) to Saathi as compensation

7.5 Security Rules

- JWT tokens expire in 15 minutes — refresh token lasts 30 days
- All wallet amounts in API responses are rounded to 2 decimal places
- KYC documents encrypted at rest — URLs are pre-signed S3 links with 1-hour expiry
- Admin actions are logged in audit_log with actor_id, action, IP, timestamp
- Rate limiting: login attempts (5 per minute), OTP resend (3 per 10 minutes)
- Sub-Admin cannot access modules not assigned to their role

8. Recommended Tech Stack

Layer	Technology	Justification
Backend API	**Node.js + Express / NestJS**	Fast development, great ecosystem for microservices
Database	**PostgreSQL**	Robust transactions, PostGIS for geo-queries, JSONB support
Cache	**Redis**	Session storage, rate limiting, service list TTL cache
Message Queue	**RabbitMQ / Kafka**	Async commission processing, event-driven architecture
File Storage	**AWS S3 / GCS**	KYC documents, profile photos — pre-signed URL access
Auth	**JWT + bcrypt**	Industry standard, stateless, easy to validate cross-service
Mobile	**React Native / Flutter**	Cross-platform iOS + Android from single codebase
Admin Panel	**React.js + Ant Design**	Rich data tables, forms, charts for admin workflows
Payment Gateway	**Razorpay / Stripe**	Wide coverage, webhook support, refund management
SMS/OTP	**Twilio / MSG91**	OTP delivery, notification SMSes for Saathi
Maps / Geo	**Google Maps API / Mapbox**	Task location display, Saathi proximity search
Monitoring	**Prometheus + Grafana**	Service health, transaction volumes, error rates
DevOps	**Docker + Kubernetes + GitHub Actions**	Container orchestration, automated deployments

8.1 Microservice Communication

- Synchronous: REST/HTTP between services for real-time requests (auth validation, user lookup)
- Asynchronous: Message queue for commission events, notifications, audit logs
- Service discovery: Kubernetes DNS or API Gateway routing
- Each service has its own database — no cross-service DB joins
- Shared data accessed via service-to-service API calls only

8.2 Environment Setup (Per Developer)

- Phase 1-2 Developers: Node.js, PostgreSQL, Redis — local Docker Compose
- Phase 3-4 Developers: Add RabbitMQ to Docker Compose + Saathi geo setup
- Phase 5-6 Developers: Business service + Partner service with country seeding
- All Developers: Shared Postman collection for API testing + seed data scripts